Limits of Resolution in Quantum Machine Learning: A Statistical Noise Budget for Small Benchmarks

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Abstract

Quantum machine learning (QML) benchmarks on small tabular datasets are usually read as rankings of architectures. We ask how large a difference such a benchmark can resolve and how much the answer depends on the inferential procedure. On a Western Mediterranean precipitation panel (1938–2026; 1-month-ahead exceedance of a regional 60th percentile; 528 test station-months), one AUC has a standard error of about 0.024, which exceeds the across-seed variability of Random Forest, XGBoost and an LSTM; additional seeds do not remove the uncertainty due to the finite test sample. For the three classical comparisons of 15-seed means, the paired minimum detectable differences (MDEs) are 0.031–0.044 AUC. Random Forest–XGBoost (+0.021) is borderline ($p = 0.054$), and Random Forest–LSTM (+0.015) is excluded from zero by year-block bootstrap intervals but not by DeLong ($p = 0.19$), so conclusions about small gaps depend on the procedure. In a case study matched on fitting rows and preprocessing, with 15 seeds per model, Random Forest exceeded a hybrid quantum model (24 quantum parameters) by 0.029 AUC (DeLong $p = 0.026$, unadjusted for multiplicity), whereas a circuit-free classical control differed from the hybrid by about 0.001 (intervals extending to ±0.012–0.017). Simulations with known differences show that the DeLong interval was approximately calibrated when year effects were shared by the scorers (type-I error 0.04) but not when they were scorer-specific (0.07–0.22), while the year-block bootstrap kept coverage near 0.92–0.93 and rejected the null slightly too often (0.07–0.08). No comparison establishes improvement over a month×station climatology. Re-evaluating fixed classical predictors under alternative label definitions, given that 23 % of the test months are incomplete, preserved the direction of the AUC differences but changed their nominal significance. We propose a reporting protocol for small QML benchmarks: a stated estimand, a noise budget with an MDE, a climatological reference and matched fitting sets.

Keywords: Quantum Machine Learning, Benchmarking, Statistical Power, Minimum Detectable Effect, Seed Variance, Precipitation Forecasting.

1 Introduction

Claims of quantum utility on classical tabular data frequently rest on a single training run per model, capacity-asymmetric classical baselines, and uncertainty estimates that cover only one source of error (Bowles et al., 2024). The machine-learning literature has long pointed out that a single run confounds initialization luck with architectural merit (Henderson et al., 2018; Bouthillier et al., 2021), and that tests computed on one fixed test set ignore the variance induced by the training procedure (Dietterich, 1998; Demšar, 2006). Variational circuits aggravate the issue: in the noisy intermediate-scale quantum (NISQ) era (Preskill, 2018), parameterized quantum circuits are sensitive to initialization and prone to barren plateaus (McClean et al., 2018), and they are usually combined with a classical backbone that compresses the input into a few qubits (Huang et al., 2021; Schuld & Killoran, 2022).

Throughout, "noise" means statistical variability (test-set sampling error and optimization-seed variance), not quantum hardware noise. The standard remedy is to repeat training with several seeds. This paper argues that, on small benchmarks, the remedy addresses the source of error that is smaller on the scale of an individual AUC: with a few hundred correlated test observations, the sampling error of each AUC is much larger than the variation caused by the seed, and the design can reliably detect only differences of a few hundredths of AUC. We also show that, at this scale, the conclusion about a small difference can depend on which inferential target and procedure are used, so these choices must be stated.

We demonstrate this on a deliberately modest but realistic case: a multi-decadal, multi-station Western Mediterranean precipitation panel, an area where heavy precipitation is linked to cut-off lows aloft (DANAs; Romero et al., 1999; Nieto et al., 2005) and where tree ensembles are strong baselines on engineered tabular features (Grinsztajn et al., 2022). The task is a 1-month-ahead probabilistic forecast of precipitation exceeding a pooled regional 60th percentile (P_{60} = 28.1 mm) from antecedent 12-month features. The panel is a testbed for evaluation methodology, not an operational early-warning system.

Applied climate- and weather-QML studies have addressed separate parts of the problem: opportunities for climate modelling (Schwabe et al., 2025a), parameter-count and initialization-matched comparisons for cloud cover (Schwabe et al., 2025b; Pastori et al., 2026), wind-speed forecasting (da Silva et al., 2026), quantum kernels for climate emulation (Bazgir & Zhang, 2024), synthetic time-series benchmarks (Fellner et al., 2026), variational data assimilation (Kumar et al., 2026), and parameter-matched neural comparisons (Marchisio et al., 2026). In the time-series and weather QML studies we read, quantum models are reported as better than classical baselines from a table of point estimates: Marchisio et al. (2026) report a 15.4 % lower MSE than a parameter-matched classical neuron on a 2,000-sample test set with one value per model and no interval or test, and da Silva et al. (2026) use 10 runs per configuration and show their spread graphically, but report no interval or test on test sets of 14 and 5 days. We do not claim that these effects are spurious; the point is that the evidence reported cannot separate them from sampling and seed noise. This paper complements these studies by asking how large a difference an evaluation design can resolve.

What the study shows. In a benchmark of 528 correlated observations, a protocol that separates test-sampling error, optimization seeds, temporal dependence and label validity changes qualitatively which conclusions can be drawn from AUC differences smaller than about 0.03; we propose and illustrate that protocol as a minimal evaluation standard for small QML benchmarks.

Contributions.

1. **A statistical framework** that separates two estimands, the difference of single-run AUCs and the difference of seed-averaged AUCs, and gives paired DeLong, year-block bootstrap and two-way (years and seeds) bootstrap inference for each, with the MDE as a design-sensitivity measure (Section 3).
2. **A noise budget** for Random Forest, XGBoost and an LSTM on 528 test station-months, comparing sampling error with seed variance (Section 4.1).
3. **Known-truth simulations** of two correlated scorers with shared and model-specific year effects, which check the type-I error, coverage and power of the procedures at this sample size (Section 4.2).
4. **An audit of label validity** for incomplete monthly totals, with the sensitivity of the labels and of the three classical comparisons to the label definition, by re-evaluating fixed predictors (Section 4.3).
5. **A QML case study and a reporting protocol**: an empirical benchmark in which a hybrid quantum model, a circuit-free classical control, a standalone LSTM and Random Forest, fitted on the same rows with 15 seeds each, are compared with the same paired procedures alongside a climatological reference (Sections 4.4 and 4.5), yielding the recommended reporting protocol collected in Section 5.2.

Further material (positive control, single-seed analyses, other hybrid architectures, ablations, operational metrics, circuit costs) is in the Supplementary Material.

2 Data, Task and Models

2.1 Dataset, Target and Split

The study domain is the Spanish Mediterranean coast (comprising Catalonia, the Valencian Community and the Region of Murcia), a transition zone between Mediterranean (*Csa*) and semi-arid (*BSk*) Köppen regimes subject to severe convective episodes (Romero et al., 1999; Nieto et al., 2005). Daily station records were compiled from historical series of the Spanish Meteorological Agency (AEMET) hosted by *datosclima.es* under the AEMET Open Data framework. The series span 1938–2026 at four stations (Barcelona, Valencia, Alicante and San Javier) and give 3,363 station-months; each station loses its first 12 months as antecedent window ($4\times12 = 48$), so the 3,363 station-months yield 3,315 prediction windows. Daily variables are aggregated to monthly resolution per station (precipitation sum, wettest day, days above the daily 90th percentile, mean pressure, temperature, diurnal range, pressure variability, wind speed, maximum gust, reference evapotranspiration (Allen et al., 1998) and sunshine duration). Months with fewer than 10 valid daily records are excluded, and monthly totals are sums over the available days, so incomplete months enter the data (Section 4.3). Inhomogeneities from relocations or instrument changes remain an accepted limitation (Alexandersson, 1986; Peterson et al., 1998).

Antecedent observations from months $i, \ldots, i+11$ (35 input features: 12 precipitation lags, 17 engineered physical features, 2 cyclic month terms and 4 station indicators) predict whether precipitation at month $i+12$ exceeds the pooled regional 60th percentile $P_{60} = 28.1$ mm, computed on the training partition only. The split is chronological: training 1938–1996 ($N = 2{,}273$ windows), validation 1997–2007 ($N = 514$)

and test 2008–2026 (N = 528; positive rate 41.1 %, 217 positives). Station records are assembled by an inner join across five daily variable files, so San Javier (pressure ends 2010) and Valencia (2012) are under-represented in the test partition: 83.7 % of test samples come from Alicante and Barcelona (Table 1). Because regimes differ between humid Barcelona/Valencia and semi-arid Alicante, part of the pooled AUC reflects base-rate differences between stations. Covariates are standardized with training-partition statistics only; re-computing the upstream monthly anomalies from pre-1997 data shifts the AUC of the three primary models by at most 0.004 (Supplement). Standardizing covariates strictly with pre-1997 statistics anchors anomalies to the cooler historical baseline; this introduces a non-stationary covariate shift in test-period temperature anomalies and derived thermal contrasts (`t_anomaly`, `choque_clausius`) during accelerated warming, making the out-of-sample split a test of robustness under distribution shift. All empirical conclusions concern the complete-variable panel retained by the processing pipeline, not the full set of station-months with precipitation observations.

Table 1: Station-month windows by partition and test-set class prevalence for the regional exceedance threshold (P_{60} = 28.1 mm).

Station	Train (1938–1996)	Val (1997–2007)	Test (2008–2026)	Total	Test positive rate
Alicante	671	132	221	1,024	29.4% (65 / 221)
Barcelona	378	132	221	731	50.2% (111 / 221)
San Javier	535	132	36	703	36.1% (13 / 36)
Valencia	689	118	50	857	56.0% (28 / 50)
Total	**2,273**	**514**	**528**	**3,315**	**41.1% (217 / 528)**

2.2 Models and Fitting Sets

We evaluate two reference baselines (a month×station climatology fitted on the training partition, and persistence), classical models (logistic regression, Random Forest with 300 trees (Breiman, 2001), gradient boosting, XGBoost with 500 estimators and depth 4, and a standalone bidirectional LSTM with 10,103 parameters), and hybrid quantum–classical models simulated noiselessly in PennyLane: `LSTM+QNN` (2,613 parameters, of which 24 quantum), `LSTM+PI-QNN`, `LSTM+QTL` (frozen LSTM, quantum transfer learning; Mari et al., 2020), `LbL-QNN` (Skolik et al., 2021) and a quantum-kernel SVM. The representative hybrid of this paper is `LSTM+QNN`: it compresses a 51-dimensional vector (16-dimensional LSTM state concatenated with the 35 engineered features) to 4 rotation angles before a 4-qubit circuit. Its controls are three classical controllers on the same 4-dimensional bottleneck (linear, 20 parameters; LeakyReLU, 28; GELU, 180). Architectures, parameter counts and hyperparameters are given in the Supplement.

The historical and primary classical experiments use different fitting-set conventions; the matched QML core experiment of Section 4.5 uses the same 2,273 fitting rows for all four models. The scikit-learn tree ensembles are fitted on training plus validation (N = 2,787 rows, 1938–2007); the neural and hybrid models use the 2,273 training rows and the 514 validation rows for early stopping. All models are scored on the same 528 test rows. To check that the asymmetry does not drive the comparison, Random Forest was also fitted on exactly the 2,273 training rows: it reaches 0.6802 ± 0.0035 AUC in the canonical pipeline, against 0.6851 ± 0.0026 with 2,787 rows; in the train-only pipeline the 2,273-row Random Forest reaches 0.6846 ± 0.0036 (Table 11) against 0.6830 ± 0.0035 with 2,787 rows, demonstrating that variations of this scale ($\sim$ 0.002–0.005) are within optimization and seed variance rather than a systematic

sample-size effect. The remaining pairwise comparisons in this paper of tree and neural models use the training-row-matched Random Forest where indicated.

Every gradient-optimized model and every tree ensemble is trained with 15 seeds; the seed index is nominal across frameworks. The primary metric is the ROC-AUC on the 528 test rows, complemented by the Brier score (Brier, 1950). The train-only pipeline, in which anomalies are standardized with pre-1997 data, is the primary one for Random Forest, XGBoost and the LSTM; `LSTM+QNN` and its circuit-free control were re-run in the train-only pipeline on the same 2,273 rows (Section 4.5); the other hybrids were fitted in the canonical pipeline and were not re-run.

3 Statistical Framework

Table 2: Map of experiments: which pipeline, fitting rows, seeds and stored probabilities stand behind each result, and its inferential status.

Experiment	Pipeline	Fitting rows	Seeds	Stored test probabilities	Use (tables)
Core QML	train-only	2,273 (all four models)	15	all seeds	primary (Tables 11 and 12)
Classical rerun	train-only	trees 2,787; LSTM 2,273	15	all seeds	primary (Tables 3 and 4)
Label sensitivity	saved predictions of the two rows above	none (no retraining)	15	all seeds	sensitivity (Tables 7 and 9)
Historical hybrids and ablations	canonical	trees 2,787 (RF also 2,273); neural 2,273	15 (AUCs)	seed 0 only, except `LSTM+QNN`	descriptive (Supplement)
Single runs	canonical	as above	1 (seed 42)	single run	descriptive (Table 10)

3.1 Estimands and Inference

For models A and B with test probabilities $p_{A,s}$, $p_{B,s}$ from seeds $s = 1, \ldots, S$ ($S = 15$), the empirical statistic of this paper is the difference between seed-averaged test AUCs,

$$\hat{\Delta} = \frac{1}{S}\sum_{s} \widehat{\mathrm{AUC}}(p_{A,s}) - \frac{1}{S}\sum_{s} \widehat{\mathrm{AUC}}(p_{B,s}), \tag{1}$$

and, for single-run comparisons, the difference of two individual AUCs. Sampling inference targets the corresponding performance difference for the evaluation population, conditional on the observed training data and on the fitted seed-specific models. A bootstrap that also resamples seeds targets the seed-averaged difference over the distribution of optimization seeds, again conditional on the training data. The mean of AUCs is linear in the DeLong structural components, so the paired sampling variance of $\hat{\Delta}$ is obtained by averaging the components over seeds; this treats the seeds as fixed.

3.2 Sampling Uncertainty

Test-set sampling error is estimated in three ways. (i) The paired DeLong estimator for correlated ROC curves (DeLong et al., 1988; Hanley & McNeil, 1982) treats station-months as independent and may therefore be optimistic for monthly data. (ii) A calendar-year block bootstrap resamples the 19 test years (2008–2026) with all stations of a year together (B = 2,000; Künsch, 1989; Obuchowski, 1997). Bootstrap confidence intervals are the 2.5th and 97.5th percentiles of the replicated statistic, and bootstrap standard errors are the sample standard deviations of the replicates; resamples that contain only one class are discarded. (iii) A two-way bootstrap resamples years and, within each model, seeds. With only 19 clusters the block bootstrap is itself imprecise and for some pairs returns a smaller standard error than DeLong; all procedures are reported, and none is assumed a priori to be the correct one for the observational panel.

3.3 Seed Variability and the Seed Share

For a model with seed standard deviation σ_{seed} and mean sampling standard error SE of one AUC we report $\sigma^2_{\text{seed}}/(\sigma^2_{\text{seed}} + \text{SE}^2)$. This is a descriptive ratio of two conditional variances (seeds, on a fixed test set; test sampling, for a fixed trained model), not a decomposition of the total uncertainty of a training procedure; it compares the scale of the two sources.

3.4 Minimum Detectable Difference

For a paired difference with standard error SE_Δ, the minimum detectable difference at two-sided $\alpha = 0.05$ and power 0.80 is

$$\text{MDE} = (z_{0.975} + z_{0.80})\,\text{SE}_\Delta \approx 2.80\,\text{SE}_\Delta. \tag{2}$$

Under a normal approximation with a locally constant standard error, the MDE describes the sensitivity of the design to a true difference: true differences below it are expected to be declared significant in fewer than 80 % of repeated experiments. It is not a threshold that an observed difference must exceed, and an observed gap between $1.96\,\text{SE}_\Delta$ and $2.80\,\text{SE}_\Delta$ can be nominally significant. We therefore report observed gaps with their intervals and use the MDE only to state which true differences the design would detect reliably.

3.5 Known-Truth Simulations and Scope of Claims

Two scorers with a known population AUC difference are simulated to check the type-I error, coverage and power of the procedures (design in Section 4.2). The noise budget, the paired comparisons, the simulation and the climatology comparison are the primary analyses of the paper, and the architecture ablations in the Supplement are secondary exploratory analyses. None was pre-registered; Holm correction is applied to the DeLong p-values within families (two families of three seed-mean comparisons, the classical models and the hybrid case; bootstrap intervals are pointwise and not multiplicity-adjusted; Holm-adjusted p-values are the formal correction, and the first-step threshold 0.017 for three comparisons is quoted in table captions for reference), and every nominal p-value outside these two families is hypothesis-generating. Seed-level tests (Wilcoxon, Friedman, paired t) compare seeds on one fixed test set, omit test sampling error, and are reported as statements about optimization variance.

4 Results

4.1 The Noise Budget

On the scale of one AUC, conditional test-sampling variability exceeds across-seed variability for all three classical models (Table 3): one AUC has a standard error of 0.023–0.027 under either estimator, whereas the seed standard deviation is 0.004–0.006, so the seed accounts for 2–7 % of the combined seed-plus-sampling variance (a descriptive comparison of scales, not an estimate of the share of total uncertainty that more seeds could remove). The hybrid quantum models have seed standard deviations of 0.009–0.017. The 15-seed `LSTM+QNN` of Section 4.5 has a seed standard deviation of 0.0152. For the other three hybrids (`LbL-QNN`, `LSTM+QTL`, `LSTM+PI-QNN`) only the seed-0 test probabilities were stored, so no seed share and no paired comparison is computed for them. On this scale the sampling term is the larger one for the models analysed.

Table 3: Noise budget for one AUC, train-only pipeline, 15 seeds per model, 528 test rows. iid: DeLong; block: calendar-year block bootstrap (B = 2,000, 19 years). Pairwise differences in Table 4 are computed from unrounded seed means.

Quantity	Random Forest	XGBoost	Standalone LSTM
Mean AUC (15 seeds)	0.683	0.662	0.668
SE of one AUC, iid / block	0.023 / 0.026	0.024 / 0.025	0.024 / 0.027
SD across seeds	0.0035	0.0054	0.0063
Seed share of combined variance	**2 %**	**5 %**	**7 %**

Table 4 compares the 15-seed means on the full 2,787-row classical baseline hierarchy (Δ = +0.015 for RF − LSTM; on the strictly matched 2,273-row training set evaluated in Section 4.5, the paired difference between RF and standalone LSTM is Δ = +0.022, 0.6846 − 0.6630). The standard error of a paired difference is about half that of one AUC, and averaging the seeds lowers it further, but the DeLong MDE remains 0.031–0.044 (block estimates are 0.034, 0.017 and 0.043). For RF − XGBoost the difference is +0.021 (z = 1.93, p = 0.054) and both bootstrap intervals include zero: weak, borderline evidence. For RF − LSTM both bootstrap intervals exclude zero by a small margin, but DeLong does not (p = 0.189), and the block standard error (0.006) is about half of the DeLong one (0.011); with 19 clusters the block estimate is itself imprecise: its MDE (0.017) is substantially smaller than the iid one, and given the limited number of clusters and the uncertainty about the dependence structure neither estimate is treated as definitive. We therefore describe this comparison as dependent on the inferential procedure and not as established or refuted. XGBoost against the LSTM shows no evidence of a difference, and none of the three DeLong comparisons rejects the null after Holm correction over the three pairs (smallest p = 0.054 against a first-step threshold of 0.017). The bootstrap intervals in Table 4 are pointwise 95 % intervals, are not multiplicity-adjusted, and no bootstrap p-values were computed.

Single-seed comparisons are less stable. The paired DeLong standard error of two single runs is 0.013–0.018 (MDE 0.036–0.052), and in the train-only pipeline RF > XGBoost is declared significant in 27 % of the 15 × 15 seed pairings (21 % under the block bootstrap), RF > LSTM in 3 % (24 %) and XGBoost versus LSTM in 0 % (Supplement). In the canonical pipeline the first comparison was significant in 66 % of pairings, a change produced by a preprocessing detail that moves the AUC by less than 0.005. The canonical single-run gap RF − LSTM was +0.035 (year-block bootstrap p = 0.001), compared with a

Table 4: Paired comparison of the 15-seed means (train-only pipeline): $\Delta = \overline{\mathrm{AUC}}_A - \overline{\mathrm{AUC}}_B$ with the paired DeLong standard error on seed-averaged components, two-sided p-value, 95 % intervals from the calendar-year block bootstrap (19 years, $B = 2{,}000$) and from a two-way bootstrap that resamples years and seeds, and the MDE (2.80× SE) under DeLong and the block bootstrap.

Pair	Δ	SE (DeLong)	z (p)	Block 95 % CI (SE)	Two-way 95 % CI	MDE iid / block
RF − XGBoost	+0.021	0.0110	1.93 (0.054)	[−0.002, +0.045] (0.0122)	[−0.003, +0.046]	0.031 / 0.034
RF − LSTM	+0.015	0.0112	1.31 (0.189)	[+0.002, +0.025] (0.0061)	[+0.001, +0.027]	0.031 / 0.017
XGBoost − LSTM	−0.007	0.0159	−0.42 (0.677)	[−0.039, +0.022] (0.0153)	[−0.037, +0.023]	0.044 / 0.043

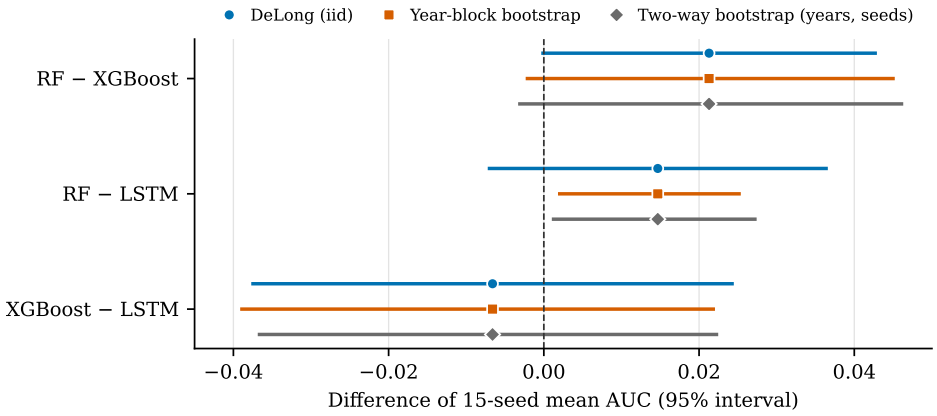


Figure 1: Differences of 15-seed mean AUC with 95 % intervals from the three procedures (train-only pipeline, 528 test rows). The procedures agree for RF − XGBoost and XGBoost − LSTM and disagree on whether the RF − LSTM interval excludes zero.

train-only seed-mean gap of +0.015. Because both the preprocessing and the training realization differ, this contrast does not isolate initialization effects.

4.2 Simulation with Known Truth

The positive control of the Supplement perturbs observed probabilities, so it cannot establish the coverage or power of the procedures over new samples. We therefore simulate two scorers whose population AUC difference is known by construction or approximated on large reference samples (four independent draws of 400,000 rows; Monte Carlo standard error of the population difference at most 0.0002). Labels have prevalence 0.41 with year-level effects on the logit; 528 rows are split among 19 years as equally as possible or, in one scenario, unequally (multinomial with Dirichlet(3) shares); scorers are binormal with noise correlation $\rho = 0.9$, which reproduces the paired DeLong standard error of the classical comparisons (0.011); the design AUCs without year effects are 0.65 and 0.65 + Δ; year effects enter as a prevalence shift (τ_π), a shift shared by both scorers (τ_s) and, in scenarios C–E, a shift specific to each scorer and independent between them (τ_m); B has $\tau_\pi = \tau_s = 0.3$, C adds $\tau_m = 0.3$, D is C with unequal year sizes, and E has all three equal to 0.6. The target is the pooled population difference for new years drawn with equal weight (details in Section A). When the design difference is 0 the scorers are exchangeable and the population difference is exactly 0. Each scenario has 500 replications; the block bootstrap uses $B = 300$ (Section A). Rejection rates have a Monte Carlo standard error of about 0.01 near 0.05.

Table 5 gives three results. First, with year effects shared by the two scorers (scenario B) DeLong is calibrated (type-I error 0.040, 20 of 500; coverage 0.960; standard deviation of the estimate 0.0108 against

Table 5: Simulation with known truth ($N = 528$, 19 years, prevalence 0.41, $\rho = 0.9$, 500 replications). τ_π, τ_s and τ_m: standard deviations of the year effects on the prevalence logit, on a shift shared by both scorers and on a shift specific to each scorer. Type-I error and coverage are for an exactly null difference (DeLong, block bootstrap); SD($\hat{\Delta}$) is the Monte Carlo standard deviation of the estimate, to be compared with the mean standard error (DeLong / block). Last column: power at the largest simulated difference (population difference in brackets). All rows are in the Supplement.

Scenario	Type-I DeLong	Type-I block	Cover. DeLong	Cover. block	SD($\hat{\Delta}$)	Mean SE	Power (DeLong / block)
B: shared year effects ($\tau_\pi = \tau_s = 0.3$)	0.040	0.074	0.960	0.926	0.0108	0.0111 / 0.0108	0.74 / 0.78 (0.029)
C: + scorer-specific year effects ($\tau_m = 0.3$)	0.068	0.076	0.932	0.924	0.0154	0.0144 / 0.0153	0.50 / 0.47 (0.028)
D: C with unequal year sizes	0.096	0.084	0.904	0.916	0.0168	0.0144 / 0.0155	0.51 / 0.48 (0.028)
E: all $\tau = 0.6$	0.220	0.074	0.780	0.926	0.0290	0.0185 / 0.0273	0.35 / 0.21 (0.023)

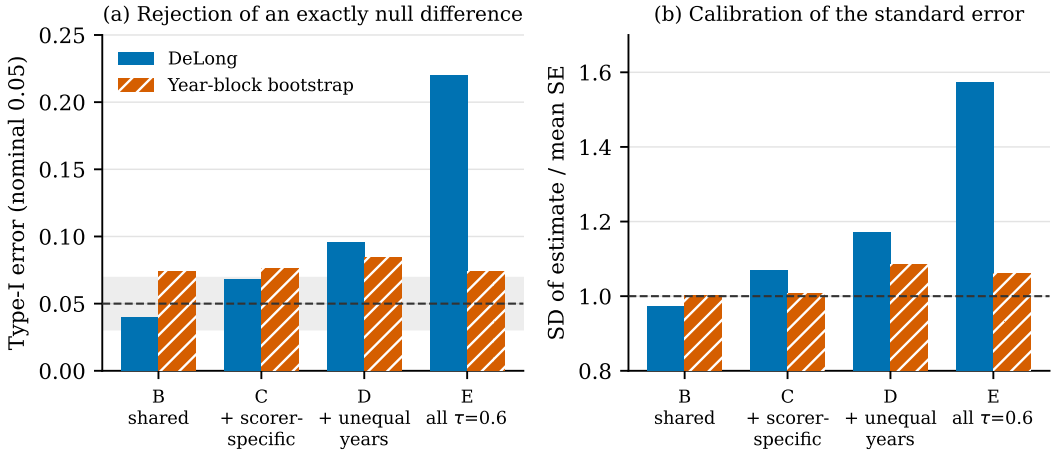


Figure 2: Simulation scenarios B–E (Table 5). (a) Rejection rate of an exactly null difference; the band is the 95 % Monte Carlo interval around the nominal 0.05 for 500 replications. (b) Ratio of the Monte Carlo standard deviation of the estimate to the mean standard error; values above 1 mean that the standard error is too small.

a mean standard error of 0.0111), whereas the block bootstrap rejects the exact null in 7.4 % of replications (Monte Carlo standard error 1.2 points) with coverage 0.926. Second, with scorer-specific year effects the DeLong standard error underestimates the standard deviation of the estimate (ratio 1.07 in C, 1.17 in D with unequal years, 1.57 in E) and its type-I error rises to 0.068, 0.096 and 0.220 with coverage 0.932, 0.904 and 0.780; the block standard error tracks the standard deviation (ratios 1.01, 1.08 and 1.06), and its type-I error stays at 0.076, 0.084 and 0.074 with coverage 0.924, 0.916 and 0.926. Neither procedure is therefore uniformly better calibrated: DeLong fails when year effects are scorer-specific, and the block bootstrap with 19 clusters is less sensitive to the scenario but liberal (type-I error 0.074–0.084, coverage 0.916–0.926), so nominal coverage was not achieved uniformly in these scenarios; additionally, in scenarios C and E at intermediate differences ($\Delta_{\text{true}} \approx 0.011$–$0.014$) the empirical mean of the finite-sample estimates showed a slight positive displacement of 0.0025–0.0040 above the reference (about 3 Monte Carlo standard errors), contributing to the undercoverage. Third, power follows the MDE rule where DeLong is calibrated: in scenario B (standard error 0.011, MDE 0.031) a true difference of 0.029 is detected in 74 % of replications (block 78 %) and 0.0145 in 32 % (35 %); in scenarios C–E the larger standard deviation of the estimate (0.015–0.029) raises the MDE accordingly. The mechanism is Gaussian and binormal with equal-weight years and no spatial structure or optimization noise; it checks the procedures at this sample size and does not reproduce the dependence of the real panel.

Panel-matched check. To assess whether the stylized scenarios misrepresent the panel, we repeated the simulation using the observed year sizes (19 years, 10–48 rows per year), the observed test prevalence (0.411), the AUCs of the seed-averaged RF and LSTM predictions (0.684 and 0.673; observed ensemble gap 0.0115, distinct from the +0.015 difference of seed-mean AUCs), and $\rho = 0.89$, estimated from their within-class normal scores. The year-level variance of both prevalence and scores was not distinguishable from sampling noise: the method-of-moments estimates were negative and were therefore truncated to zero. We therefore varied a common year-effect scale τ from 0 to 0.3 (prevalence-logit and shared-score shifts with standard deviation τ, scorer-specific shifts with $\tau/2$), with 500 replications per value. Across this grid, the type-I error of the paired DeLong test was 0.034–0.056 (coverage 0.944–0.966), whereas that of the calendar-year block bootstrap was 0.054–0.092 (coverage 0.908–0.946). The mean DeLong and block standard errors, 0.0117–0.0123 and 0.0113–0.0125, were close to the empirical standard deviation of $\hat{\Delta}$, 0.0109–0.0132. These standard errors imply an approximate 80 %-power MDE of 0.033. At this ensemble RF−LSTM gap of 0.0115 the estimated power was only 0.14–0.19 for DeLong and 0.17–0.22 for the block bootstrap; a difference of that size would thus be detected in fewer than 25 % of replications at this sample size. As in the other scenarios, this check cannot identify the year effects present in the real panel and omits spatial structure and optimization noise.

Conditional perturbation (positive control). In the Supplement, the test probabilities of Random Forest are degraded by calibrated logit noise until the AUC falls by a known amount, and the detection rate of the undegraded model is recorded. The 80 % detection rate is reached at an imposed AUC reduction of about 0.04 and 17–30 % of runs detect a reduction of 0.02, consistent with the DeLong MDE of single-seed pairings. This is a perturbation of observed probabilities, not a simulation from a population, and it does not establish coverage or power over new samples.

4.3 Label Validity under Incomplete Months

Monthly precipitation is the sum of the available daily values, and months with at least 10 valid days are retained. Of the 3,363 retained station-months, 327 (9.7 %) are incomplete (fewer valid days than days in the month), and the share is much higher in the test partition (123 of 528, 23.3 %; 7.5 % in training) and in Barcelona (217 of 743 months, 29.2 %). An incomplete month can only under-count precipitation, so negatives may be over-represented among incomplete months. Table 6 recomputes P_{60} and the labels under stricter criteria. Requiring 15, 20 or 25 valid days changes 0, 3 and 8 labels among the common months (0, 0 and 2 in test); keeping complete months only changes 17 labels (2 in test) but removes 123 of the 528 test rows, so this check mixes label changes with a change in test composition. A proportional-coverage scenario, which scales each monthly sum by days in month over valid days, assumes that missing days rain like observed days; it is a sensitivity scenario and neither a validated correction nor a bound. It moves the threshold from 28.0 to 28.4 mm, changes 32 labels (14 in test) and raises the test prevalence to 0.430; at the project threshold, 18 incomplete months (12 in test, 2.3 % of test rows) change label between the plain and the corrected sum.

Table 6: Sensitivity of the monthly labels to the coverage criterion. Flips are labels that differ from the project labels among the common months (in brackets: test partition). Prevalence and n refer to the test partition.

Criterion	Months kept	P_{60}	Flips (test)	Test prevalence	Test n
Project: ≥ 10 days, plain sum	3,363	28.0	–	0.411	528
≥ 15 days, plain sum	3,354	28.1	0 (0)	0.413	523
≥ 20 days, plain sum	3,336	28.1	3 (0)	0.416	512
≥ 25 days, plain sum	3,284	28.3	8 (2)	0.417	489
Complete months only	3,036	27.3	17 (2)	0.407	405
≥ 10 days, proportional-coverage scenario	3,363	28.4	32 (14)	0.430	528

The audit recomputes P_{60} from the training station-months of the monthly table (28.0 mm) and not from the windowed dataset (28.1 mm); the labels obtained with 28.0 mm agree with the dataset labels on all 528 test rows, so this difference changes no test label.

Observed rain discarded by the join. For the 3,363 station-months of the panel, we compared the precipitation summed over the days retained by the inner join across the five daily variable files with the sum over all days with observed precipitation in the precipitation-only files (the latter is not a complete monthly total when days are also missing there). Of these station-months, 325 lose at least one day (1,299 days in total) and 121 (3.6 %) lose observed rain, 1,776.6 mm in total; among the affected months the median loss is 6.8 mm, the 95th percentile 55.4 mm and the maximum 125.4 mm. The loss is concentrated in Barcelona (83 of the 121 months) and in the test partition (56 months, 666.2 mm; Valencia's 50 test months are unaffected). At the fixed project threshold 20 labels change (9 in test, 11 in training, none in validation); with the threshold recomputed from the unjoined training sums (28.6 mm instead of 28.0 mm) 41 labels change (12 in test). A further 897 station-months with at least 10 days of observed precipitation do not enter the panel because fewer than 10 days survive the join; this is a selection of months and not a change of labels, and its effect was not examined.

Whether the model comparisons change is examined with fixed predictors and without retraining (scenarios S0–S7; Tables 7 to 9). Three kinds of re-evaluation are reported: evaluation on a subset of rows with the

Table 7: Re-evaluation of the saved test predictions (15-seed means, no retraining) under alternative label definitions: sample, labels and scores of the classical models. S4–S7 are hypothetical scenarios that evaluate predictors trained on the project labels against alternative labels. Pos.: positive test labels. Flips: test labels that differ from the project labels. Differences between models are in Tables 8 and 9.

Scenario	n	Pos.	Prev.	Flips	AUC RF / XGB / LSTM	Brier RF / XGB / LSTM
S0 project labels, all rows	528	217	0.411	0	0.683 / 0.662 / 0.668	0.224 / 0.246 / 0.239
S1 complete target months	405	163	0.402	0	0.695 / 0.681 / 0.689	0.219 / 0.234 / 0.228
S2 $\geq$ 25 valid days	489	206	0.421	0	0.695 / 0.678 / 0.681	0.221 / 0.238 / 0.233
S3 ambiguous labels dropped	516	217	0.421	0	0.688 / 0.667 / 0.676	0.222 / 0.243 / 0.236
S4 coverage-corrected, same threshold	528	229	0.434	12	0.688 / 0.667 / 0.678	0.223 / 0.243 / 0.234
S5 coverage-corrected, new threshold	528	227	0.430	14	0.690 / 0.670 / 0.680	0.222 / 0.242 / 0.233
S6 unjoined sum, same threshold	528	226	0.428	9	0.686 / 0.660 / 0.679	0.223 / 0.246 / 0.234
S7 unjoined sum, new threshold	528	221	0.419	12	0.684 / 0.661 / 0.679	0.223 / 0.246 / 0.235

project labels (S1–S3), evaluation of all rows against labels from the proportional-coverage sum (S4, S5), and against labels from the unjoined sum (S6, S7). Retraining under another label definition or pipeline was not done. The predictions are those of the train-only pipeline, 15 seeds per model; the hybrid case includes the 15-seed re-run of `LSTM+QNN` and its comparators (Table 9), and the other hybrids are not included.

For the classical models the sign of every paired difference is the same in all eight scenarios (RF − XGBoost +0.014 to +0.026; RF − LSTM +0.005 to +0.015; XGBoost − LSTM −0.003 to −0.019), but their nominal significance is not. RF − XGBoost has DeLong p between 0.019 and 0.285, below 0.05 only in S6 and S7 (unjoined-sum labels), and a year-block interval that excludes zero only in S6 ($p = 0.019$, not below the Holm first-step threshold of 0.017 for three pairs). RF − LSTM has p between 0.189 and 0.622, never below 0.05, and, among the eight scenarios, a year-block interval that excludes zero only under the original labels, S0; with unjoined-sum labels its difference falls to +0.005–+0.007. The year-block intervals come from one bootstrap run ($B = 2000$); a second run with a different random-number stream moved the endpoints of the classical pairs by up to 0.0027, and the lower limit of the S0 interval of RF − LSTM is +0.0006 here and +0.0018 in Table 4. Both are positive, but the margin by which this interval excludes zero is smaller than the run-to-run variation of its endpoints, so the exclusion is narrow and should not be interpreted as robust evidence. The comparisons that are borderline in the main analysis thus move in opposite directions under plausible label definitions, and none reaches nominal significance in a way that is stable across them. Restricting to complete months raises the AUC of the three models by 0.01–0.02 but reduces the test size to 405 rows. The Brier scores keep the same order (Random Forest lowest) in all scenarios.

For the hybrid case (Table 9), the RF − `LSTM+QNN` difference keeps its sign (+0.014 to +0.029) but not its nominal significance: the DeLong p ranges over 0.026–0.312, is below 0.05 only in S0, S2 and S3 and never below 0.017; the year-block interval excludes zero in S0 and S2–S5 and includes it in S1 (complete months) and in S6 and S7 (unjoined-sum labels). The Linear-4D control differs from the hybrid by at most 0.003 in every scenario (year-block intervals from −0.018 to +0.014), and the standalone LSTM by +0.004 to +0.009 with intervals that include zero in all scenarios.

Table 8: Paired differences of the classical models under alternative label definitions (Δ, seed-mean AUC difference) with year-block 95 % interval and DeLong p (seed-averaged components, unadjusted). The year-block intervals come from one bootstrap run ($B = 2000$) of the same script; a second run with a different random-number stream moved interval endpoints of these pairs by up to 0.0027, and the S0 interval of RF − LSTM differs from Table 4 (lower limit +0.0006 against +0.0018).

Scenario	RF − XGB: Δ [block CI]; p	RF − LSTM: Δ [block CI]; p
S0 project labels, all rows	+0.021 [−0.002, +0.045]; 0.054	+0.015 [+0.001, +0.026]; 0.189
S1 complete target months	+0.014 [−0.012, +0.041]; 0.285	+0.006 [−0.010, +0.024]; 0.617
S2 ≥ 25 valid days	+0.018 [−0.005, +0.041]; 0.112	+0.015 [−0.001, +0.029]; 0.191
S3 ambiguous labels dropped	+0.021 [−0.003, +0.044]; 0.054	+0.012 [−0.0002, +0.023]; 0.265
S4 coverage-corrected, same threshold	+0.021 [−0.003, +0.043]; 0.052	+0.010 [−0.003, +0.023]; 0.358
S5 coverage-corrected, new threshold	+0.020 [−0.003, +0.043]; 0.064	+0.010 [−0.003, +0.022]; 0.375
S6 unjoined sum, same threshold	+0.026 [+0.002, +0.049]; 0.019	+0.007 [−0.009, +0.021]; 0.541
S7 unjoined sum, new threshold	+0.023 [−0.001, +0.048]; 0.035	+0.005 [−0.010, +0.020]; 0.622

Table 9: The hybrid case (Section 4.5) under the same label definitions: AUC of Random Forest (2,273 rows), standalone LSTM, Linear-4D control and `LSTM+QNN`, and paired differences with `LSTM+QNN` (Δ [year-block 95 % CI]; unadjusted DeLong p). Holm first-step threshold for these three pairs: 0.017.

Scenario	AUC RF / LSTM / Linear-4D / QNN	RF − QNN	Linear-4D − QNN	LSTM − QNN
S0 project labels, all rows	0.685 / 0.663 / 0.656 / 0.655	+0.029 [+0.012, +0.045]; 0.026	+0.001 [−0.013, +0.012]; 0.917	+0.008 [−0.001, +0.017]; 0.200
S1 complete target months	0.695 / 0.686 / 0.678 / 0.681	+0.014 [−0.009, +0.038]; 0.312	−0.003 [−0.018, +0.011]; 0.701	+0.004 [−0.005, +0.015]; 0.504
S2 ≥ 25 valid days	0.696 / 0.676 / 0.671 / 0.671	+0.026 [+0.004, +0.047]; 0.049	+0.000 [−0.014, +0.013]; 0.951	+0.006 [−0.004, +0.015]; 0.368
S3 ambiguous labels dropped	0.689 / 0.671 / 0.663 / 0.663	+0.027 [+0.009, +0.044]; 0.041	+0.001 [−0.012, +0.012]; 0.904	+0.008 [−0.001, +0.018]; 0.173
S4 coverage-corrected, same threshold	0.689 / 0.673 / 0.666 / 0.665	+0.024 [+0.005, +0.044]; 0.069	+0.001 [−0.013, +0.013]; 0.900	+0.009 [−0.001, +0.018]; 0.153
S5 coverage-corrected, new threshold	0.691 / 0.675 / 0.668 / 0.668	+0.023 [+0.004, +0.042]; 0.079	+0.000 [−0.013, +0.012]; 0.959	+0.008 [−0.003, +0.018]; 0.194
S6 unjoined sum, same threshold	0.688 / 0.675 / 0.670 / 0.668	+0.020 [−0.001, +0.040]; 0.133	+0.002 [−0.011, +0.014]; 0.791	+0.007 [−0.003, +0.017]; 0.250
S7 unjoined sum, new threshold	0.686 / 0.674 / 0.669 / 0.668	+0.017 [−0.003, +0.037]; 0.194	+0.000 [−0.012, +0.011]; 0.947	+0.006 [−0.004, +0.015]; 0.367

4.4 Reference Baselines: Climatology and Probabilistic Skill

A month×station climatology fitted on the training partition reaches AUC 0.680 (Brier score 0.2224; Brier skill score +0.082 relative to the training base rate of 0.4017, that is, to the Brier score 0.2422 of the constant forecast 0.4017 evaluated on the test rows, whose prevalence is 0.411); persistence is weaker (lag-1 precipitation AUC 0.589). Random Forest exceeds the climatology by +0.003 AUC (range −0.002 to +0.011 across seeds) and no seed is significantly better under either test; XGBoost and the LSTM score below the climatology on point estimates (−0.018 and −0.011), also without significance. On proper scoring (Table 10), none of the ten single-run models improves on the climatological Brier score. Random Forest comes closest (0.2230; BSS −0.003); the hybrid models have a high expected calibration error (ECE 0.112–0.149; its direction was not analysed), with skill scores of −0.058 to −0.140.

Table 10: Single-run AUC and Brier score of selected models against the month×station climatology (canonical single run, seed 42; 528 test rows; the full set of ten models is in the Supplement). $\mathrm{BSS}_{\mathrm{clim}} = 1 - \mathrm{BS}/0.2224$.

Model	**AUC**	**Brier score**	**$\mathrm{BSS}_{\mathrm{clim}}$**
Month×Station Climatology	0.680	0.2224	0
Random Forest	0.688	0.2230	−0.003
Gradient Boosting	0.671	0.2236	−0.005
Logistic Regression	0.655	0.2258	−0.015
LbL-QNN	0.672	0.2352	−0.058
LSTM+QNN	0.639	0.2536	−0.140

4.5 Case Study: A Hybrid Quantum Model under the Noise Budget

The core experiment fits, in the train-only pipeline and on the same 2,273 training rows (the validation rows are used only for early stopping), Random Forest, the standalone LSTM, a classical control with the architecture of the hybrid in which the 4-qubit circuit is replaced by a linear $4 \to 4$ map (20 parameters), and the hybrid `LSTM+QNN` (24 quantum parameters), with 15 seeds each and the test probabilities of every seed stored. Table 11 gives the seed means. The four models share the 2,273 fitting rows, the validation rows for early stopping (gradient-trained models), the 528 test rows and the 15 seed indices, which are nominal across frameworks; sharing rows does not imply equal capacity, tuning budget or input representation: the hyperparameters, feature sets, stopping rules, seeds and output files of each model are given in Table S3 of the Supplement and defined in `qml_core_trainonly.py`. Random Forest has the highest mean; the hybrid has the lowest and a seed standard deviation about twice that of the classical neural models. The paired comparisons of the seed means use the procedures of Section 4.1 (Table 12).

Table 11: 15-seed test AUC (train-only pipeline, same 2,273 fitting rows; mean and sample SD), sorted by mean. The last column is the difference to `LSTM+QNN`; it is descriptive, not a test. Climatology: AUC 0.6796.

Model	Mean AUC	SD	Gap to LSTM+QNN
Random Forest (2,273 rows)	0.6846	0.0036	+0.029
Standalone LSTM	0.6630	0.0067	+0.008
Linear-4D classical control	0.6559	0.0074	+0.001
LSTM+QNN (24 quantum parameters)	0.6552	0.0152	0

Table 12: Paired comparisons of the 15-seed means with `LSTM+QNN` (train-only pipeline, 528 test rows, 19 test years; same procedures and notation as Table 4). DeLong p-values are unadjusted; the Holm first-step threshold for these three pairs is 0.017. The MDEs are specific to each pair.

Difference	$\hat{\Delta}$	SE (DeLong)	z (p)	Year-block 95 % CI (SE)	Years×seeds 95 % CI	MDE DeLong / block
RF − LSTM+QNN	+0.029	0.0132	2.22 (0.026)	[+0.012, +0.045] (0.0084)	[+0.010, +0.048]	0.037 / 0.024
Linear-4D − LSTM+QNN	+0.001	0.0066	0.10 (0.917)	[−0.012, +0.012] (0.0062)	[−0.016, +0.017]	0.019 / 0.017
LSTM − LSTM+QNN	+0.008	0.0061	1.28 (0.200)	[−0.002, +0.017] (0.0047)	[−0.005, +0.021]	0.017 / 0.013

Random Forest exceeds the hybrid by 0.029 AUC. At the nominal 5 % level without multiplicity adjustment, DeLong ($p = 0.026$) and both bootstrap intervals (year-block [+0.012, +0.045]; years×seeds [+0.010, +0.048]) exclude zero, but the DeLong difference does not survive Holm correction across the three comparisons of this family (adjusted $p = 0.079$, computed from unrounded p-values; first-step threshold 0.017), and the bootstrap intervals are pointwise and not multiplicity-adjusted. The estimate is imprecise (the intervals span about 0.04). Under the alternative label definitions of Section 4.3 (Table 9) the difference stays between +0.014 and +0.029 but its nominal significance varies. The hybrid is not distinguished from the circuit-free classical control: the difference is +0.001 and the year-block and years×seeds intervals extend to ±0.012 and ±0.017, so the data are compatible with differences between the 15-seed means from −0.012 to +0.012 (year-block) and from −0.016 to +0.017 (years×seeds); no equivalence margin was specified in advance, and this is not an equivalence test. Against the standalone LSTM the difference is +0.008 with intervals that include zero (in this core experiment the standalone LSTM scored 0.6630 ± 0.0067, compared with 0.6682 ± 0.0063 in Table 3, reflecting seed-level shuffle and initialization variance across independent training runs). A bottleneck-free LSTM with the same hidden size, two further 4-dimensional controllers with other activations and full-feature XGBoost, evaluated

in the canonical pipeline (Supplement), score 0.6435–0.6575. These experiments do not establish the four-dimensional bottleneck as the cause of the observed Random Forest–hybrid gap. The quantum parameters are 0.9 % of the trainable parameters of `LSTM+QNN`, so most trainable parameters are classical, although this ratio alone does not quantify the functional contribution of the circuit. For `LSTM+QTL`, whose pre-trained extractor is frozen, the extractor alone scores 0.6657 ± 0.0091 against 0.6519 ± 0.0128 for the hybrid; only the seed-0 probabilities of the other hybrids (`LbL-QNN`, `LSTM+QTL`, `LSTM+PI-QNN`) were archived, so no paired interval of seed means is available for them and neither a degradation nor an improvement over the classical extractor can be claimed.

5 Discussion

5.1 What a Benchmark of This Size Can and Cannot Conclude

Under the iid DeLong approximation the estimated 80 %-power MDE for the three classical seed-mean comparisons on 528 correlated test months is 0.031–0.044 AUC; block-based estimates differ, particularly for RF − LSTM (0.017). The observed gaps between the architectures studied (0.003–0.03) are mostly smaller than these values. This does not mean that the models are equal. Three observations follow. First, for the individual-AUC scale comparison reported here, conditional test-sampling variability exceeded across-seed variability (seeds are 2–7 % of the combined variance of the classical models); this is a comparison of scales and not an estimate of how much of the total uncertainty additional seeds could remove. Second, for gaps of this size the conclusion depends on the estimand and on the procedure: single-seed verdicts change with the pipeline (RF > XGBoost declared significant in 66 % of seed pairings with the canonical preprocessing and 27 % after a change that moves the AUC by less than 0.005), and for the seed means DeLong and the bootstrap disagree on RF − LSTM. Third, the reference points are strong: the observed mean RF–climatology AUC difference is about 0.003, no single-run model improves the climatological Brier score, and the reported comparisons do not establish skill beyond the seasonal-geographic base rate, whatever its architecture.

For the quantum results, `LSTM+QNN` scores 0.029 AUC below Random Forest fitted on the same rows (DeLong $p = 0.026$, not surviving correction over three pairs; year-block interval +0.012 to +0.045; across the label scenarios of Section 4.3 the difference stays between +0.014 and +0.029 with p from 0.026 to 0.312) and is not distinguished from a circuit-free classical control (difference +0.001; intervals extend to about ±0.012–0.017); the calibration of the 15-seed hybrid was not assessed in this analysis. The available evidence does not establish a reproducible quantum performance advantage in this case study, and uncertainty remains substantial for differences of the magnitude observed here; the results neither show nor rule out an advantage on larger or richer data.

5.2 Recommended Reporting Protocol

Table 13 lists the practices that follow from the results. Items 1, 2 and 4 need only the test probabilities of the models; item 3 needs a specified and executed generating mechanism.

Table 13: Reporting protocol for small QML benchmarks.

#	Practice
1	State the estimand (single run or seed average; conditional on which sources) and report paired confidence intervals for it, with DeLong and a block bootstrap over the natural cluster (here calendar years); report discrepancies between procedures instead of selecting one.
2	Report the MDE as a measure of design sensitivity next to the intervals; do not treat it as a significance threshold or classify gaps below it as uninformative.
3	Check the procedures against a simulation with known differences at the sample size and dependence structure of the study.
4	Report across-seed variability separately from test-sampling uncertainty; if a relative-scale ratio is used, state its conditional and descriptive interpretation, and treat seed-level tests as statements about optimization, not generalization.
5	Include a climatological (or base-rate) reference with a proper scoring rule, and report skill relative to it.
6	Fit all model families on the same rows and tune them under a comparable budget, or state and quantify the asymmetry.
7	Add a classical control at the same low-dimensional interface as the quantum circuit.
8	Audit label validity (here, incomplete months) and re-evaluate the comparisons under alternative labels.
9	Choose equivalence margins by scientific relevance, independently of the MDE and of the results, include sampling error, and treat an interval wider than the margin as inconclusive.

5.3 Limitations

Inferential procedures. The block bootstrap relies on 19 calendar years and, for RF − LSTM, gives about half the DeLong standard error; the iid estimator ignores temporal dependence. The simulations (Section 4.2) show how the procedures behave under stylized mechanisms (Gaussian binormal scores, shared and scorer-specific year effects, no spatial structure, no optimization noise): DeLong was anticonservative when year effects were scorer-specific, and the block bootstrap was somewhat liberal; the discrepancy observed for RF − LSTM (block standard error smaller than DeLong) is in the opposite direction to scenarios C–E, where the block standard error exceeded or tracked DeLong, whereas in the panel-matched check both estimators were closely aligned (0.011–0.012). In clustered panels with a small number of clusters ($K = 19$), a block standard error below the iid value can arise from negative inter-cluster correlation across years or small-sample cluster dispersion, which is why we treat the discrepancy as inferential sensitivity rather than definitive proof of variance reduction. The simulations do not establish which procedure is best calibrated for the observational panel, so both are reported and their discrepancy is treated as inferential sensitivity. Block lengths other than one calendar year were not examined, and 2026 is a partial year.

No retraining under alternative labels. The label scenarios re-evaluate fixed predictors; whether the Random Forest–hybrid gap would persist if the models were retrained under another label definition was not examined.

Hybrid models are only partly covered. Only `LSTM+QNN` has 15-seed archived predictions; for `LbL-QNN`, `LSTM+QTL` and `LSTM+PI-QNN` only the seed-0 probabilities were archived, so seed-averaged paired intervals and MDEs are not available for them. The `LSTM+QNN` comparisons concern one ansatz in noiseless simulation, and no equivalence margin was specified in advance.

Label validity. Incomplete months are 23 % of the test partition (Section 4.3). The models were not retrained under the alternative label definitions, months with incomplete coverage among the 12 antecedent

months of a window were not examined, and 897 station-months with at least 10 observed precipitation days do not enter the panel because of the inner join and were not examined.

Baseline tuning and scope. Hyperparameters were set by validation-based manual grids; a tuned Random Forest on 2,273 rows (0.6798 ± 0.0045) scores slightly below the untuned one on the same rows, so tuning moves the AUC by less than the noise, but XGBoost (0.662) should not be read as a ceiling for boosted trees. The panel is small (3,363 station-months), monthly aggregation does not resolve convective episodes, and the chronological split falls in a period of accelerated warming, so with one chronological split we cannot determine how distribution shift modifies model performance or sensitivity to optimization randomness. We evaluate one ansatz family in noiseless simulation with a 0.9–3.9 % quantum share of trainable parameters; inference-time gate noise changes the results little (Supplement) but says nothing about noise during training (Wang et al., 2021).

Generality beyond this dataset. All empirical conclusions concern the complete-variable panel retained by the processing pipeline, not the full set of station-months with precipitation observations (897 station-months with at least 10 observed days do not enter it; Section 4.3), and the evidence is empirical for one panel, but the sampling-error argument does not depend on the climate application. For a binary classifier with AUC A, n_1 positives and n_0 negatives, the Hanley–McNeil approximation (Hanley & McNeil, 1982) gives the standard error of one AUC from (A, n_1, n_0) alone; for the present test set it returns 0.024, matching the DeLong value. Assuming a correlation of 0.75 between the AUCs of two models scored on the same rows (the paired standard errors measured here imply 0.72–0.83), Table 14 gives the approximate 80 %-power MDE for other test-set sizes: about 0.049 for $N = 500$ at prevalence 0.41 and about 0.086 at prevalence 0.10. These values are specific to the assumed AUC (0.68), inter-model correlation and iid approximation; an MDE is a design sensitivity and not a bound on detectable effects, and temporal and spatial dependence can change it in either direction for paired comparisons (for RF − LSTM the block standard error here is smaller than the DeLong one). They do not replace validation on further datasets.

Table 14: Approximate MDE for the AUC difference of two models on the same test set (Hanley–McNeil standard error, AUC 0.68, inter-model correlation 0.75, $\alpha = 0.05$, power 0.80). Test-set size N and positive-class prevalence π.

N	MDE, $\pi = 0.41$	MDE, $\pi = 0.10$
250	0.069	0.122
500	0.049	0.086
1,000	0.035	0.061
2,500	0.022	0.038
5,000	0.015	0.027

6 Conclusions

On a benchmark of 528 correlated test station-months, conditional test-sampling variability exceeded across-seed variability on the scale of one AUC; under the iid DeLong approximation the estimated 80 %-power MDE for the three classical seed-mean comparisons was 0.031–0.044 AUC, with block-based estimates differing, particularly for RF − LSTM; and conclusions about gaps of one to two hundredths depend on the inferential procedure: Random Forest against XGBoost is borderline, and Random Forest against the LSTM is excluded from zero by year-block bootstrap intervals but not by DeLong. In the

simulated scenarios, DeLong inference was approximately calibrated with shared year effects and became anticonservative as scorer-specific year effects increased; the block bootstrap was less sensitive to the simulated scenario but not nominal (type-I error 0.074–0.084, coverage 0.916–0.926). Neither procedure is selected a priori, and their discrepancy is reported. The reported comparisons do not establish improvement over the climatology, no single-run model improves its Brier score, and the circuit-free classical control is within +0.001 AUC of the quantum hybrid (intervals extending to about ±0.012–0.017), while Random Forest exceeds the hybrid by 0.029 (DeLong $p = 0.026$, not surviving correction), a difference whose nominal significance varies across the label definitions (DeLong p from 0.026 to 0.312). The defensible conclusion is not that Random Forest is superior, but that this benchmark cannot resolve differences of this size reliably. About 23 % of the test months are incomplete. Re-evaluating the saved predictions of the three classical models on subsets of months and against alternative labels kept the sign of every paired difference but not its nominal significance: among the eight label scenarios the year-block interval of RF − LSTM excludes zero only under the original labels, and that exclusion is narrow and should not be interpreted as robust evidence, while RF − XGBoost reached $p = 0.019$ only with labels from the unjoined precipitation sum. No model was retrained under alternative labels.

The consequence for QML benchmarking is practical. Single-seed comparisons, and multi-seed comparisons that ignore sampling error, can report differences that the benchmark cannot reliably support, and the inferential target and procedure must be stated. Reporting the estimand, the noise budget with its MDE, a simulation check, a climatological reference and a label audit makes the evidential scope of a small benchmark explicit and separates claims the data can carry from claims they cannot.

Data and Code Availability

Code, processed panel data and model checkpoints are available at https://github.com/enriquerb32/limits-of-resolution-qml. The canonical pipeline is anchored in checkpoint `01_data.pt` (monthly panel $N = 3{,}363$, 1938-03 to 2026-05; windowed dataset $N = 3{,}315$; train/validation/test 2,273/514/528; 35 input features); thresholds ($P_{60} = 28.1$ mm, $P_{90} = 86.32$ mm) were computed on the training partition only. The daily station records are sourced from AEMET's public data holdings via *datosclima.es* under AEMET's Open Data reuse framework; only the processed, aggregated monthly panel is redistributed (`results/panel_checkpoint.csv`), with attribution to AEMET. Scripts for the analyses of this paper are organized under `code/local_experiments/` (`exp7_noise_budget.py`, `exp10_paired_ci.py`, `exp11_simulation_known_truth.py`, `exp12_monthly_coverage_audit.py`, `exp13_label_sensitivity.py`, `exp14_innerjoin_audit.py`, `exp15_panel_matched_simulation.py`); figures are generated from saved results via `scripts/make_paper_figs.py`. The 15-seed QML core experiment (Table 11) is produced by `code/local_experiments/qml_core_trainonly.py` (outputs `results_qml_core_trainonly.json` and `results_qml_core_trainonly_probs.npz`); the paired comparisons of Table 12 are produced by `exp10_paired_ci.py` with the option `--extra` pointing to that archive, and the label sensitivity of the hybrid (Table 9) by `exp13_label_sensitivity.py` with the same option; test metadata are exported by `export_test_meta.py`. The commands, seeds and output files are listed in the repository README. Internal variable names (`dana_*`, `acp_*`) and the feature glossary are in the Supplement.

Declaration of Generative AI in Scientific Writing

During the preparation of this manuscript, the author used generative AI tooling for editorial language polishing, LaTeX formatting and structural drafting, and for drafting analysis scripts that the author ran and checked. No reported test statistic, empirical metric or benchmark result was generated by AI models; the numerical results come from the released scripts. The author assumes full scientific and editorial responsibility for the content.

Author Contributions

E.R. conceived the study, curated the data, developed the models, performed the statistical analysis, and wrote the manuscript.

Competing Interests

The author declares no competing financial or non-financial interests.

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A Computations

Estimators. For two models scored on the same 528 rows, the DeLong estimator gives the covariance of the two AUCs from their structural components (DeLong et al., 1988); it treats station-months as independent. The block bootstrap resamples the 19 calendar years (2008–2026) with replacement, keeping all stations of a year together, recomputes both AUCs on each resample ($B = 2{,}000$) and takes the standard deviation of the paired difference as its standard error (Künsch, 1989). With 19 clusters the estimate of the standard error is imprecise, and the direction of its discrepancy with the DeLong value depends also on the dependence structure (here the block standard error is below the DeLong value for RF − LSTM: 0.010 versus 0.014 for single seeds, 0.006 versus 0.011 for seed means). The two-way bootstrap draws a calendar-year resample and, independently for each model, a resample of its 15 seeds, and recomputes the difference of seed-averaged AUCs.

Seed-mean statistic. The AUC of one model is the mean of the DeLong structural components over positives and over negatives, so the mean of the 15 AUCs equals the AUC statistic built from components

averaged over seeds, and the paired DeLong variance of $\hat{\Delta}$ follows from the averaged components. This treats the 15 seeds as fixed and so conditions on the fitted models; the two-way bootstrap relaxes this by resampling seeds.

Seed share of variance. For model m with seed standard deviation s_m and sampling standard error SE_m, the share is $s_m^2/(s_m^2 + \mathrm{SE}_m^2)$. Using the mean DeLong SE over the 15 seeds, Random Forest (SD 0.0035, SE 0.0232), XGBoost (0.0054, 0.0236) and the LSTM (0.0063, 0.0237) give 2.3 %, 4.9 % and 6.6 % (2 %, 5 % and 7 % in Table 3). No seed shares are computed for the hybrids: for `LbL-QNN`, `LSTM+QTL` and `LSTM+PI-QNN` only the seed-0 test probabilities were stored.

Simulation mechanism. For each replication, years $u = 1, \ldots, 19$ receive a prevalence effect $\tau_\pi\, \eta_u$ on the logit of $\pi = 0.41$ and a number of rows of 528 split as equally as possible (or, in the unequal scenario, a multinomial split with Dirichlet(3) shares). Scorers $k \in \{A, B\}$ are $s_k = \mu_k y + e_k + \tau_s\, w_u + \tau_m\, v_{k,u}$, with $e_k = \sqrt{\rho}\, z + \sqrt{1-\rho}\, \varepsilon_k$ (so that the scorer noise has correlation ρ), w_u a year shift shared by both scorers and $v_{k,u}$ a year shift specific to each scorer; all $\eta_u, w_u, v_{k,u}, z, \varepsilon_k$ are standard normal and $\mu_k = \sqrt{2}\,\Phi^{-1}(\mathrm{AUC}_k)$ without year effects. The design AUCs without year effects are 0.65 for scorer B and $0.65 + \Delta$ for scorer A; the scorer noise has correlation $\rho = 0.9$, and the scorer-specific year shifts $v_{k,u}$ are independent between scorers. Scenario B has $\tau_\pi = \tau_s = 0.3$ and $\tau_m = 0$; C adds $\tau_m = 0.3$; D is C with Dirichlet year sizes; E has $\tau_\pi = \tau_s = \tau_m = 0.6$. The simulation target is the pooled population difference under the mechanism, with years drawn independently and weighted equally, that is, the difference for new years; in scenario D the shares of the 528 rows among years follow a Dirichlet(3) draw that is independent of the year effects, the sample statistic pools observations (so it weights years by their sizes), and the reference is the pooled population AUC with equally likely years of equal size. Scenario D changes the distribution of cluster sizes. Because the sizes are independent of the year effects, the pooled statistic is expected to be centred on the equal-weight reference; this was checked empirically and not proved: in D the mean of the estimates differs from the reference by +0.0002, −0.0008 and −0.0004 for design differences of 0, 0.015 and 0.03 (Monte Carlo standard error of the mean about 0.0008). The population AUC of each scorer is computed as the mean of four independent draws of 400,000 rows from the same mechanism with years drawn independently and equally weighted, and the Monte Carlo standard error of the population difference is reported. When the design difference is 0 the scorers are exchangeable and the population difference is set to 0. DeLong tests use the paired estimator of the seed-free case; the block bootstrap resamples years ($B = 300$ per replication).

Label re-evaluation (script `exp13_label_sensitivity.py`). Saved test predictions are linked to the monthly coverage table through the station and target month of each test window. Scenarios S0–S7 are the project labels on all rows; complete target months; target months with at least 25 valid days; rows with unambiguous labels (the plain and the proportional-coverage sums agree at the project threshold); proportional-coverage labels at the same threshold and at the recomputed threshold; and labels from the unjoined precipitation sum (script `exp14_innerjoin_audit.py`) at the project threshold and at the threshold recomputed from the unjoined training sums. S4–S7 evaluate predictors trained on the project labels against alternative labels; they do not evaluate a pipeline retrained under those labels.